

Hongyi Pan, Ph.D.

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🔗 <https://www.scholar.google.com/citations?user=23I1CMYAAAAJ> in hongyi-pan-67a2b1181

Education

University of Illinois Chicago

PhD in Electrical and Computer Engineering, Advisor: Ahmet Enis Cetin.

Aug 2019 – Aug 2023

Chicago, IL, USA

- Concentration in AI-based wildfire detection, anomaly detection, and efficient neural networks.
- Dissertation: Orthogonal Transform Domain Based Neural Network Layers.

University of Illinois Chicago

MS in Electrical and Computer Engineering

Aug 2017 – May 2019

Chicago, IL, USA

Chang'an University

BS in Automation

Sep 2014 – July 2018

Xi'an, Shaanxi, China

Experience

Postdoctoral Research Fellow

Northwestern University, Advisor: Ulas Bagci

June 2023 – Present

Chicago, IL, USA

- Lead researcher on federated foundation models for medical image analysis.
- Directed the curation and benchmarking of multiple large-scale medical image datasets, including **Cyst-X** for pancreatic analysis and **LUMINA** for multi-vendor mammography.
- Designed a federated domain generalization framework using wavelet-regularized Fourier transforms to reduce data heterogeneity in federated learning.

Engineering Intern

InnoPeak Technology (OPPO US)

May 2022 – Aug 2022

Bellevue, WA, USA

- Developed attention-based variable rate super-sampling (VRSS) models for graphics rendering.
- Contributed to efficient deep learning models for real-time visual processing.

Skills

Programming Languages: Python, MATLAB, C, C++, LaTeX.

Scientific Tools/Software: PyTorch, TensorFlow, OpenCV, Docker, Microsoft Office, AWS, Photoshop, ...

Operating Systems: Windows, Linux, Mac OS.

Microcontrollers: Raspberry Pi, Nvidia Jetson Nano, Arduino, STM32, MSP430, 80C51.

Selected Projects

LUMINA Mammography Benchmark with Energy Harmonization (CVPR2026)

- Curated a diverse full-field digital mammography (FFDM) dataset of 1,824 images across 6 distinct acquisition systems with pathology, BI-RADS, and breast-density annotations.
- Developed a training-free, model-agnostic foreground-only pixel-space CDF alignment method to eliminate severe vendor- and energy-induced domain shifts while preserving lesion morphology.
- Benchmarked multi-view CNN and Vision Transformer architectures across three clinical tasks, achieving a state-of-the-art 93.54% AUC for pathology-confirmed diagnosis and improving Grad-CAM attention focus.

Cyst-X: Federated AI for Pancreatic Cancer Risk Prediction (NC Major Revision)

- Created and led the first large-scale, publicly available, multicenter MRI dataset project for pancreatic cystic lesion analysis, involving 7 hospitals.
- Achieved state-of-the-art performance in identifying high-risk IPMNs under privacy constraints.

Orthogonal Transform-Based Efficient Deep Neural Networks (ICML2023, TNNLS2024, IJCAI2026)

- Implemented a hybrid Quantum-Classical Hadamard neural network layer. (ICML2023)

- Implemented a series of efficient neural network layers based on biorthogonal block wavelet transform, discrete cosine transform, and Hadamard transform for convolutional neural networks. (TNNLS2024)
- Implemented a structure-preserving initialization strategy and a frequency-domain compression protocol using discrete cosine transform for Vision Transformers. (IJCAI2026)
- Improved performance and efficiency of CNNs and ViTs on various datasets such as ImageNet-1K and COCO.

VideoAds: Benchmarking MLLMs on Fast-Paced Video Understanding (ICCV2025)

- Developed VideoAds, the first dedicated VQA benchmark dataset designed to evaluate Multi-modal Large Language Models (MLLMs) on high-density commercial video structures.
- Formulated a novel quantitative video complexity density metric utilizing DINO-v2 feature variance to evaluate multi-stage temporal coherence, rapid scene transitions, and long-range semantic progression.
- Benchmarked leading open-source and proprietary MLLMs (including Qwen2.5-VL and Gemini 2.5 Pro) across visual summary and causal inference tasks.

Federated Domain Generalization for Medical Image Analysis (ICASSP2024, ICASSP2025)

- Developed federated domain generalization methods based on the Fourier Transform.
- Improved the performance on various datasets, including breast, polyp, and retina fundus.

Wildfire Detection on Edge Devices (TECS 2022)

[Demo](#) 

- Developed a MobileNetV3-based real-time wildfire detection model optimized for edge deployment.
- Successfully deployed the system on Nvidia Jetson Nano using HPWREN surveillance data.

Selected Publications

Published 30 peer-reviewed conferences and 21 journals. Google Scholar citation: 900+; h-index: 17; i10-index: 23. Full publication list: <https://www.scholar.google.com/citations?user=23I1CMYAAAAJ>

H. Pan, et al. Discrete Cosine Transform Based Decorrelated Attention for Vision Transformers. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2026.

H. Pan, et al. LUMINA: A Multi-Vendor Mammography Benchmark with Energy Harmonization Protocol. *IEEE conference on computer vision and pattern recognition (CVPR)*, 2026.

H. Pan, et al. Cyst-X: A Federated AI System Outperforms Clinical Guidelines to Detect Pancreatic Cancer Precursors and Reduce Unnecessary Surgery. *Nature Communications*, under review, major revision.

H. Pan, et al. Frequency-Based Federated Domain Generalization for Polyp Segmentation. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

Z. Zhang, M. Dou, L. Peng, **H. Pan**, U. Bagci, B. Gong. VideoAds for Fast-Paced Video Understanding: Where Opensource Foundation Models Beat GPT-4o & Gemini-1.5 Pro. *International Conference on Computer Vision (ICCV)*, 2025.

H. Pan, et al. Multichannel Orthogonal Transform-Based Perceptron Layers for Efficient ResNets. *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2024.

H. Pan, et al. Domain Generalization with Fourier Transform and Soft Thresholding. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

B. Wang, **H. Pan**, et al. GazeGNN: A Gaze-Guided Graph Neural Network for Chest X-Ray Classification. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

H. Pan, et al. A hybrid quantum-classical approach based on the Hadamard transform for the convolutional layer. *International Conference on Machine Learning (ICML)*, 2023.

H. Pan, et al. Real-time wireless ecg-derived respiration rate estimation using an autoencoder with a dct layer. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

M. Yang, Z. Ye, **H. Pan**, M. Farhat, A.E. Cetin, P.Y. Chen. Electromagnetically unclonable functions generated by non-Hermitian absorber-emitter. *Science Advances*, 2023.

H. Pan, et al. Block walsh-hadamard transform-based binary layers in deep neural networks. *ACM Transactions on Embedded Computing Systems (TECS)*, 2022

H. Pan, et al. Multiplication-avoiding variant of power iteration with applications. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022

Editorial Services

- Associate Editor, Signal, Image and Video Processing (SIVP), 2023–Present